Number Theory — problem 6.140

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Integers a, b, c, d, e are such that $n \mid a+b+c+d+e$, $n \mid a^2+b^2+c^2+d^2+e^2$ for the odd integer n. Prove that $n \mid a^5+b^5+c^5+d^5+e^5-5abcde$.